Question 1: Expression Evaluation with Short-Circuit Logic

Evaluate each of the following expressions if a is 6, b is 9, c is 14, and flag is 1. Which parts of these expressions are *not evaluated* due to short-circuit evaluation?

1. `c == a + b || !flag`
2. `a != 7 && flag || c >= 6`
3. `!(b <= 12) && a % 2 == 0`
4. `!(a > 5 || c < a + b)`

① c == a+b || !flag

14 == 6+9 || !1

14 == 15 || 0

0 || 0

0 "

False "

② a != 7 && flag || c >= 6

6 != 7 && 1 || 14 >= 6

1 && 1 || 1

1 && 1

1 "

True "

③ !(b <= 12) && a % 2 == 0

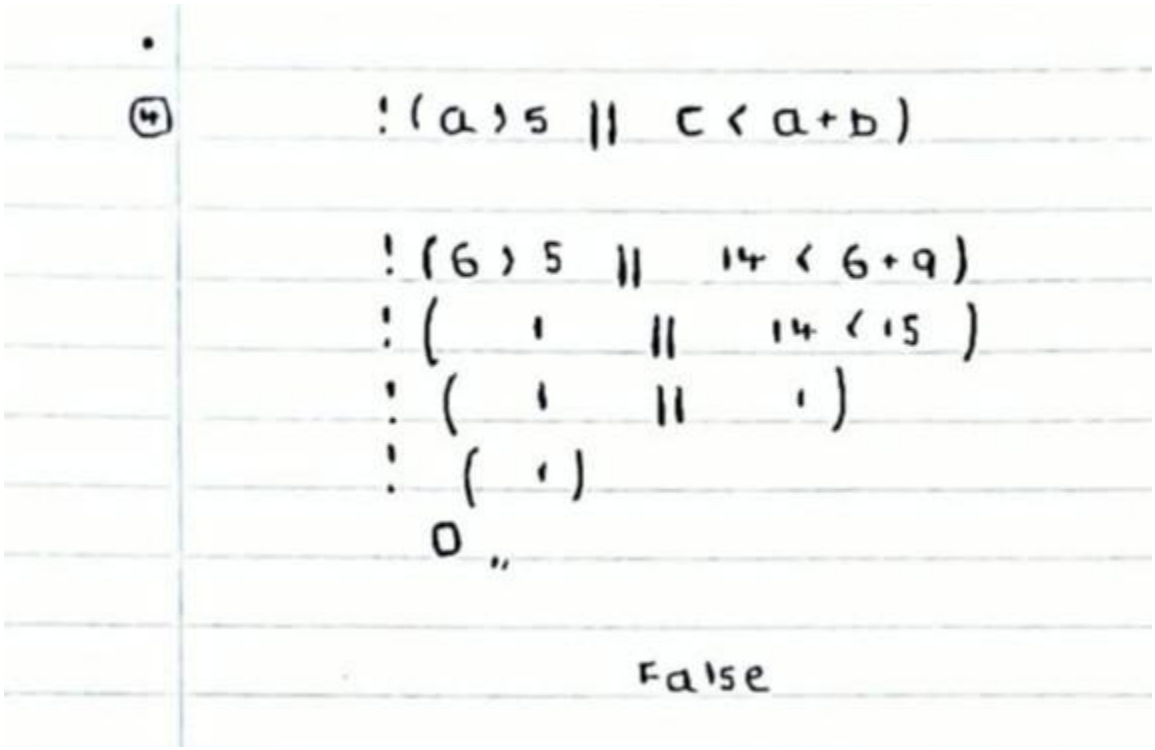
!(9 <= 12) && 6 % 2 == 0

!1 && 0 == 0

0 && 1

0 "

False


(4) !(a>5 || c<a+b)
!(6>5 || 14<6+9)
!(1 || 14<15)
!(1 || 1)
!(1)
0
False